

**MGT 449::TOPICS IN OPERATIONS & TECHNOLOGY:
GLOBAL SUPPLY CHAIN MANAGEMENT - NEW APPROACHES
SUMMER 2025**

Faculty Christopher Gopal, PhD

SESSIONS	Remote: Via Zoom: Details to be announced.
CLASS HOURS	Tuesdays August 5, 12, 19, 26; September 2, 630-920 pm Final Assignment Submission/Deliverable: Sep 5, 7 pm.
OFFICE HOURS	By Appointment
SECTION ID	890379
EMAIL	cgopal@ucsd.edu
PHONE	(858) 750-5527
TA:	Aarushi Gaikwad
EMAIL:	aagaikwad@ucsd.edu

COURSE DESCRIPTION:

Supply/Value Chains are among the most discussed topics of today at the business, societal and policy levels. In the highly competitive, volatile, uncertain, complex, and ambiguous (VUCA) international environment that we operate in, companies do not compete. *Supply chains and industries compete. And countries compete through their supply chains.* It is imperative, therefore, that managers, policy makers and students understand global supply chains, their operations, the key decisions and trade-offs and the events that impact them.

The course addresses the new issues and environment of the Global Supply Chain, the impacts of technology and people, and the new imperatives and practices for companies and governments. It is a critical course for executives, managers, policy-makers and students in all the major functions of business and policy across all industrial sectors.

The supply chain is the most significant and critical component of business and national security, and accounts for the bulk of the material, people, cash and value-added in an organization. Recent events – trade wars and tariffs, the pandemic, wars, major international crises and extreme weather events have highlighted the critical role that the supply chains play in business, global trade, the world economy, as well as in national security and societal well-being.

The supply chain involves the flow of materials, cash, information and processes from the source of raw materials to end customers... and back. Its objectives are multiple and complex and include maximizing shareholder value (and stakeholder interests) while minimizing total

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costs, investment and working capital and optimizing the customer life cycle experience. And doing all this while minimizing risk, maximizing resilience and ensuring supply chain sustainability. This “holistic”, multi-faceted and “End-to-End” view of the supply/value chain incorporates Critical Thinking and is very different from the traditional functional approach. It is geared towards understanding the modern supply chain in the modern environment.

The course will be taught using a combination of several learning approaches – classroom lectures, case analyses and projects. It will provide an overview of supply chains and their key functions, the economic environment, the financial impacts and levers, while addressing the major challenges in managing an efficient supply chain.

COURSE PREREQUISITES: This is a Master’s Level Program.

GRADING AND ASSIGNMENTS:

Lecture + Case + Research/Assignment format :65% Research/Case work + 25 % Final assignment, with 10% based on Attendance and Participation.

COURSE LEARNING OUTCOMES

Upon completing this course, students will obtain an understanding of:

- the End-to-End (E2E) supply chain, the critical component and set of functions in global business.
- the economic and business drivers of supply chain strategy and structure
- demand management, the needs of customers and the customer life cycle experience (CX)
- supply and sourcing imperatives and strategies, and matching demand and supply
- leading practices, core components and strategic aspects of today’s supply chains, including risk, resilience, and sustainability, and their drivers and enablers– technology and talent.
- the key decisions, trade-offs that must be made in today’s global, fragile, uncertain, and technologically driven supply chains.

The course will prepare students to leave this class armed with an understanding of today’s issues, what’s needed, impacts and decisions, and to be able to confidently articulate these with C-level executives and governmental leaders.

REQUIRED TEXT AND MATERIALS:

Text:

- *“Breakthrough Supply Chains: How Companies And Nations Can Thrive And Prosper In An Uncertain World”*, McGraw-Hill, 2023. ISBN: 978-1-264-98966-9; e-ISBN: 978-1-264-99188-4.

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- *“Operations Management Reading: Managing Inventory”*, Janice H Hammond, Harvard Business School, Publication Date: Sep 2013 (Revised July 11, 2018, Product Number: 8016-PDF-ENG
- *“A new tariff paradigm: How businesses can respond”*, Grant Thornton, April 14, 2025
Tariffs and Global Supply Chains: Will be provided on Canvas

Case Study:

- **“Apple’s Response to COVID-19 Disruptions: Supply Chain Risk Analysis and Mitigation”** Author: Parminder Singh Kang, Rajbir Singh Bhatti, Aditi Patil Pub. Date: 2023; DOI: <https://doi.org/10.4135/9781529610635> Publishing Company: SAGE Publications: SAGE Business Cases Originals City: London; Online ISBN: 9781529610635

Articles and news items will also be provided on Canvas to highlight current events, concepts/ideas to discuss, and context for the course.

TEAMS FOR CASE STUDIES AND TEAM ASSIGNMENT

Teams, teamwork and collaboration are an essential part of the learning process. Hence, while teams will initially be self-selected, it may be necessary for us to add team members or create new teams, if necessary. Teams can meet, collaborate and coordinate remotely. If so, it is preferable that you use Zoom for this.

CLASS ASSIGNMENTS

For certain class sessions, I shall provide a set of guiding questions, discussion points, and goals for the lecture. These will be given in the previous session. You are expected to prepare answers to these questions and be able to provide substantiated arguments on discussion points. I suggest that you work closely with your team on these assignments to share and discuss individual ideas so that you come prepared to class and contribute to a richer learning experience. It is required that you read the assigned readings before attending the class. This will help give you a perspective on the topics to be covered.

THE USE OF AI TOOLS

The use of AI tools such as GROK and ChatGPT is permitted in this class for the purposes of enhancing learning. However, it is the responsibility of each student to

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ensure that their use of AI tools does not constitute academic misconduct, such as plagiarism or cheating. It is each student's responsibility to verify the content generated by AI tools for accuracy. Students are expected to use AI tools in a responsible and ethical manner and to abide by the guidelines set forth by the instructor. Any violation of these guidelines will result in disciplinary action.

ACADEMIC INTEGRITY

Integrity of scholarship is essential for an academic community. As members of the Rady School, we pledge ourselves to uphold the highest ethical standards. The University expects that both faculty and students will honor this principle and in so doing protect the validity of university intellectual work. For students, this means that all academic work will be done by the individual to whom it is assigned, without unauthorized aid of any kind.

The complete UCSD Policy on Integrity of Scholarship can be viewed at:

<http://academicintegrity.ucsd.edu/>

How the Honor Code applies to this course:

The Final Exam is an individual assignment. However, it is a good idea to work with your Team in preparing for the exam.

ASSIGNMENTS

The following describes the guidelines for individual, collaborative or group work:

I – Independent, individual work only. No collaboration or consultation allowed during the Exam.

SCHEDULE

IMPORTANT ASSIGNMENT DATES AND TIMES

Assignments	Date/Time
Team Case Study/Assignment: Phase 1 Write-up	Aug 19th
Team Project/Assignment Phase 2: Document	Aug 28th
Final Assignment Submission/Deliverable (Notes I)	Sept 5th

All Study teams will be finalized by August 14th.

GRADING

Component	Points [or percentage]
Team Project/Assignment: Phase 1 Write-up/Discussion	15
Team Project/Assignment: Phase 2 Document/Presentation	50
Attendance/Participation	10
Final Assignment Submission/Deliverable	25
Total	100

Final grades will be based on your individual and team performance. Two items are considered when assigning final grades:

1. The overall percentage score.
2. Your ranking among all students in the class.

Class Attendance & Participation

Attendance will be taken at each class. Participation points will be awarded based on classroom session and presentation participation.

It is expected that all students will attend all the classes and participate. If, for some reason, you cannot attend a particular class, please let your teams, the TA and me know in advance.

The following are the perspectives that are critical to learning for you and your colleagues, and for good class participation: You should:

- Articulate the issues and point of view clearly.
- Make points and comments that are relevant to the class materials and discussion.
- Ask questions that are relevant to the global supply chain business environment.
- Raise interesting developments that may have occurred and that would be interesting to the class.
- Display, when discussing case studies or situations, evidence of analysis of the problem,
- Put forward new ideas, when appropriate.

The Critical aspect here is that you participate, regardless of whether you feel the points you are making, or the questions you are asking, are good.

STUDENTS WITH DISABILITIES

A student who has a disability or special need and requires an accommodation in order to have equal access to the classroom must register with the Office for Students with Disabilities (OSD). The OSD will determine what accommodation(s) may be made and provide the necessary documentation to present to the instructor and OSD liaison. Students requesting accommodations for this course due to a disability must provide a current Authorization for Accommodation (AFA) letter (paper or electronic) issued by the OSD (<https://osd.ucsd.edu/students/registering.html>).

Students are required to discuss accommodation arrangements with instructors and OSD liaisons in the department **72 business hours in advance** of any exams or assignments. **No accommodations can be implemented retroactively.**

Please visit the [OSD website](#) for further information or contact the Office for Students with Disabilities at (858) 534-4382 or osd@ucsd.edu.

CLASS AGENDA:

Please note that the topics covered in any specific class are subject to change in terms of emphasis, depth or sequence.

Session	Date	Topic(s)	Readings	Assignments Due
1	Aug 5	- Introduction to the End-to-End Supply Chain(s), understanding and definition - Evolution of the supply chain – economic, financial, trade environment and policies that have determined the current structure and management of today's supply chain.	Chapters 1 & 2	Team formation
2	Aug 12	- Customers and the Customer Experience Life Cycle (CX)– today and tomorrow - The Integrated Supply Chain & CX - Supply – Sourcing, Strategies , Challenges - Linking Demand and Supply – Demand: Supply & Inventory Deployment	Chapters 3 & 4 <i>Operations Management</i> <i>Reading: Managing Inventory</i>	Teams finalized and/or allocated Team Project/Assignments
Aug 14: Team Project/Assignment Selection				
3	Aug 19	- Demand: Supply & Inventory (contd..) - Information, Data, Governance & Analytics - Technology and the Supply Chain - Knowledge, Business Acumen, Finance and Costs	Chapters 5 & 6 <i>"A new tariff paradigm: How businesses can respond",</i>	Final Assignment outline and discussion
Aug 19:: Team Project/Assignment - Phase # 1 Write-Up Due COB				
4	Aug 26	- Supply Chain Risk Management & Mitigation - Sustainability and Supply Chains	Chapters 7 & 8	Team Project/Assignment Discussion/ Presentations
Aug 26:: Team Project/Assignment - Phase # 2 Document due before class				
5	Sep 2	- Sustainability and Supply Chains (contd) - Measures that Matter. - Conclusion	Chapters 9 & 10	Team Project/Assignment Discussion/ Presentations
6	Sep 5	Final Assignment Submission /Deliverable: 7- pm		

Faculty:

Christopher Gopal, PhD.

Dr. Gopal has over 40 years of experience in global supply chain & operations strategy, execution, and technology in a career that has encompassed industry executive management, consulting and executive education. Dr. Gopal has held executive positions at several leading companies, including Partner & Director of Global Supply Chain & Operations Services at Ernst & Young Consulting, VP in World-Wide Operations at Dell Computer, as well as executive VP positions at Unisys, Overland Storage and SAIC. His consulting clients have included prominent global and regional companies across a range of industries, and he has also assisted in several start-up companies.

Dr. Gopal has served, for several years, as a Member of the Defense Business Board (DBB), which provides the Secretary and Deputy Secretary of Defense, as well as other senior leaders, trusted independent and objective advice which reflects an outside private sector perspective on proven and effective best business practices for consideration and potential application to the Department.

Chris is also a Supply Chain Advisor to the Global Fund based in Geneva, a public-private organization designed to accelerate the end of AIDS, tuberculosis, malaria and COVID as epidemics. He has also served on a White House sub-committee on Manufacturing Technologies. Chris currently consults with companies, serves as the Strategic Advisor at several small companies, teaches at the University of California San Diego and at the University of Southern California, where he also serves on the Advisory Board of the Global Supply Chain Management Center.

Dr. Gopal has spoken and served as an invited panelist at major industry forums, including the Milken Institute Global Forum, IATA World Conferences, the OMI-OECD Forum, the Harvard Business Review Forum, the USC “Learning from the Pandemic to Improve Healthcare” Symposiums in 2021-22, the Energy Institute, Texas on “Sustainable Supply Chains”, the EU Conference on Monopoly and Security held recently in Brussels, the Panel on “*The Next World System: Can US Trade Policy Make Americans More Secure, Democratic, and Prosperous?*” at the National Press Club in Washington, and recently testified before the Congressional US-China Committee on Supply Chain security in Washington.

Dr. Gopal is the co-author of four books on Supply Chain Management. His latest book, “*Breakthrough Supply Chains: How companies and nations can thrive in an Uncertain World*” was published June 2023 and has been endorsed by the Council for Supply Chain Management Professionals and been accepted into the Pentagon Library.

Chris holds a PhD from the University of Southern California, and an MBA from the Cranfield School of Management, UK.